Labour Supply Responses to Marginal Tax Rates, Average Tax Rates, and Tax Progressivity [1]

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Abstract

Standard models of labour supply predict that individuals respond to the marginal tax rate, which determines the net-of-tax return to work at the margin. Empirical evidence, however, is mixed: some studies find strong responses to marginal incentives, whereas others find behavior more consistent with broader measures of tax burden. This paper studies labour supply responses to changes in marginal and average tax rates induced by shifts between linear and nonlinear tax schedules. We design a controlled laboratory experiment in which participants complete a real-effort task across multiple stages, constructing individualized linear and nonlinear tax schedules around each participant's prior labour supply to separately identify the effects of marginal and average tax rates. We find that participants significantly reduce labour supply under the progressive tax schedule when the average tax rate is held constant, whereas responses are less consistent

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when the marginal tax rate is held constant. This inconsistency is driven by heterogeneity between participants who are sensitive to marginal incentives and those who are not. These findings suggest that future research may benefit from explicitly accounting for distinct behavioral types.

Keywords: *Labour Supply, Taxation*

4.1 Introduction

In standard models of labor supply and taxation, individuals choose labor supply based on the net-of-tax wage on the margin, implying that the marginal tax rate governs substitution incentives. At the same time, tax systems affect after-tax income, generating income effects that may reinforce or offset these substitution effects. Empirical evidence provides mixed support for the prediction that labor supply responds to marginal incentives. On the one hand, a large literature documents bunching at kink points and notches in nonlinear tax schedules, which is commonly interpreted as evidence that individuals understand and respond to marginal incentives (Saez, 2010; Kleven and Waseem, 2013; Kleven, 2016). By contrast, a separate empirical literature finds that labour supply responses are not driven primarily by changes in marginal tax rates. Instead, observed behavior appears to be more closely associated with broader measures of after-tax income than with the true marginal incentive (König et al., 1995; Dokko, 2008; Messacar, 2024). Related work argues that optimization and adjustment frictions can substantially attenuate observed responses to tax changes, implying that even when marginal incentives are the relevant object in theory, observed behavior may reflect a combination of marginal incentives and total income that are difficult to disentangle (Chetty, 2012; Gelber et al., 2020).

A natural explanation for this mixed evidence is that individuals simplify complex tax systems and rely on more salient or cognitively accessible objects. The tax salience and tax perception literatures emphasize that taxpayers often misperceive nonlinear schedules, smooth them into simpler perceived schedules, or respond to tax-inclusive prices and total liabilities rather than to marginal incentives (Liebman and Zeckhauser, 2004; Chetty et al., 2009; Hayashi et al., 2013; Abeler and Jäger, 2015; Taubinsky and Rees-Jones, 2018; Rees-Jones and Taubinsky, 2020). A related behavioral literature on reference dependence and income targeting reaches a similar conclusion in a different context: workers may base labour supply decisions on salient earnings levels or income

targets rather than on the local return to additional work (Camerer et al., 1997; Farber, 2005, 2008; Crawford and Meng, 2011). Taken together, these literature suggest that labour supply may respond not only to marginal incentives, but also to overall after-tax income. In addition, these responses may vary with the complexity of the tax schedule, particularly whether the schedule is linear or nonlinear. Consequently, observed responses to taxation may reflect a combination of effects arising from marginal incentives, from changes in total tax liability, and from complexity and adjustment frictions, making it difficult to disentangle the underlying mechanisms in nonexperimental settings.

In this paper, we design a controlled laboratory experiment to identify the marginal and average disincentives created by taxation. Participants complete a real-effort task across multiple stages under different tax regimes. In the first stage, individuals face a linear compensation scheme with no taxation, which allows us to observe each participant's baseline productivity and willingness to work. In the second stage, depending on the treatment, we introduce either a flat or a progressive tax schedule, and participants again choose their optimal labour supply. In the third stage, we construct individualized tax schedules around each participant's stage-two labour supply. Depending on the treatment, these schedules are either progressive or flat.

In two treatments, the stage-three tax schedules are calibrated so that the average tax rate at the participant's revealed optimal effort bracket is identical to that in stage two, while the marginal tax rate differs. Because after-tax income at the stage-two labour supply is held fixed, these treatments isolate the labour supply response to changes in the marginal incentive to supply additional labour, relative to the participant's stage-two choice, while holding constant the average tax burden, that is, overall income from completing the task. In the remaining treatments, we instead calibrate the tax schedule to match the marginal tax rate at that same labour supply while allowing inframarginal taxation, and therefore the average tax rate, to vary. Because the marginal incentive to work is held fixed while after-tax income changes, these treatments identify the labour supply response to changes in the average tax rate.

By comparing labour supply across these treatments, we evaluate the extent to which labour supply responds to marginal incentives versus changes in overall after-tax income when tax schedules shift between linear and nonlinear forms. Because the tax schedules are individualized around each participant's revealed labour supply, these comparisons are made at a common reference point where the relevant components are controlled.

Our study builds on the literature that examines behavioral responses to taxation in controlled settings. Closest in spirit to our question is de Bartolome (1995), who studies how individuals respond to taxation in an investment setting and shows that many subjects behave as if the average tax rate applies to marginal decisions. de Bartolome (1995) highlights that individuals may not respond to marginal incentives in the way standard theory predicts, their environment is a static investment choice rather than a real-effort labor supply decision, and therefore does not involve a tradeoff between income and costly effort. As a result, it does not capture the joint tradeoffs that arise in labour supply settings where taxation simultaneously affects both the marginal incentives and overall after-tax income.

Methodologically, using a within-subject design to study labour response to taxation is related to Dickinson (1999) and Sillamaa (1999), who also use endogenous labor supply choices and calibrate experimental environments around participants' revealed preferences. Dickinson (1999) uses this approach to estimate the income effect (IE) and the compensated substitution effect (SE) of a wage change, finding that the compensated SE is negative, as theory predicts. Our study identifies these channels in a different way. Rather than using lump-sum transfers to control for the income effect, we employ nonlinear tax schedules to hold after-tax income constant at each participant's previously chosen labour supply. This allows us to isolate the labour supply response to marginal incentives while keeping the average tax burden fixed at the reference point. Sillamaa (1999) extends this approach to the case of nonlinear taxation by implementing a Hausman-equivalent linear tax schedule and comparing labour supply responses to wage changes under progressive and flat tax systems. The results indicate stronger responses under progressive taxation. While this design allows for comparing labour responses to wage difference between linear and progressive tax schedules, changes in progressivity simultaneously affect both marginal incentives and infra-marginal tax burdens. As a result, the design does not separately identify effects from changing the marginal tax rate and the average tax rate, as they both vary jointly away from the reference point.

We build our ATR-controlled treatments on Collins et al. (1992), Sillamaa (1997), and Carpenter et al. (2016). This literature provides theoretical predictions and experimental evidence on how labour supply responds to changes in tax progressivity in real-effort settings, generally finding that greater progressivity reduces effort, although the effects are not statistically significant in some contexts. To our knowledge, this is the first study to provide statistically significant evidence of this effect using a within-subject design with individualized ATR control and a reversed tax-schedule order. Moreover, by

inducing participants to stop working before the end of the work stage (Blumkin et al., 2012), our design allows us to examine heterogeneous labour-supply responses among participants who are sensitive to the marginal income–labour-cost tradeoff and those who are not.

Our experiment builds on these papers by using individualized nonlinear and linear tax schedules to separately manipulate the marginal return to labour and the overall tax burden at a common reference level of labor supply. By holding fixed after-tax income in one treatment and the marginal income in another, we isolate the marginal and average disincentives created by taxation. This allows us to provide a relatively clean test of how labor supply responds to changes in marginal incentives versus changes in overall after-tax income in a controlled setting.

4.2 Experiment

4.2.1 Procedure

The experiment was conducted at McDSL, which contains 22 computerized stations. All tasks were programmed in oTree (Chen et al., 2016). Upon arrival, participants were randomly assigned to cubicles. A general message and the experimental instructions were displayed on-screen and also read aloud to all participants. Following the main instructions, which are common to all treatments (Section 4.6.1), participants completed a comprehension quiz to confirm understanding of the task. Earnings from the task were paid privately in cash at the end of the session.

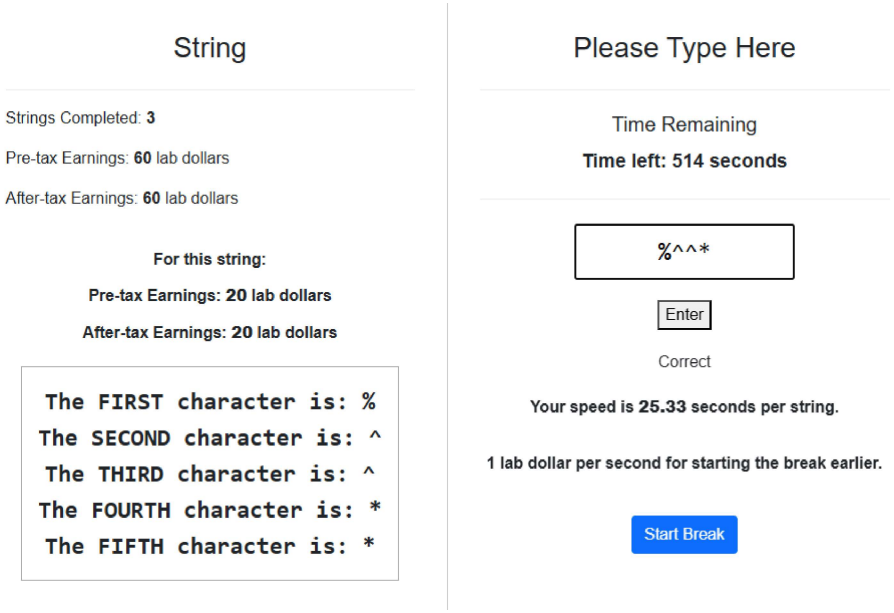
4.2.2 Task and Design

The experiment consisted of three ten-minute work shifts on a typing task, separated by two five-minute breaks. The first work shift was untaxed in order to familiarize participants with the task environment and to provide baseline measures of both productivity (Seconds per string) and willingness to work. The task requires participants to type a string of five special characters into a text box. These characters correspond to the symbols produced by pressing the number keys 1 through 9 while holding the Shift key. The interface displays the number of strings completed, pre-tax earnings, after-tax earnings for the current shift, and the before-tax and after-tax earnings associated with completing the current string.

A countdown timer indicates the time remaining in the current work shift. The interface also displays the participant's average completion speed for strings completed during that shift (Figure 4.1). In addition, the interface includes a "Start Break" button, which allows participants to begin their break before the end of a ten-minute work shift. Choosing to end a work shift early simply transfers the remaining work time to the subsequent break period; thus, doing so changes only the allocation of time between work and break. During breaks, participants are allowed to use their phones or browse the internet. Experimental earnings are denominated in lab dollars, with an exchange rate of 1 lab dollar = CAD 0.005. Each correctly completed string earns 20 lab dollars before tax, and each second reallocated from work time to break time earns 1 lab dollar.

Depending on the treatment, the second and third work shifts differ in the order of flat (linear) and increasing (progressive) tax schedules. The tax schedule in each subsequent shift is calibrated to match either the average tax rate or the marginal tax rate from the bracket corresponding to the participant's labour supply in the previous shift (see section 4.8 for details). This generates a 2×2 mixed design with four treatments: *Prog-Flat-ATR*, *Flat-Prog-ATR*, *Prog-Flat-MTR*, and *Flat-Prog-MTR*. The first two labels denote the sequence of tax schedules, while the final label indicates whether the average tax rate or the marginal tax rate is held constant.

FIGURE 4.1: Screenshot of the Typing Task



Notes: The typing task was consistent across the treatments. The information regarding earnings was on the left side of the given string. The information regarding the time and speed was on the right side with the text box, the enter button and the start break button.

4.2.3 Participants

Participants were recruited from the existing Mestelman & Muller McMaster Decision Science Laboratory (McDSL) participant pool using the SONA management system. The experiment consisted of four treatments: *Prog-Flat-ATR*, *Flat-Prog-ATR* , *Prog-Flat-MTR*, and *Flat-Prog-MTR*. The total sample size is to be 153 participants, 39, 36, 40, and 38 respectively assigned to each treatment. On average, participants received approximately $23.45 CAD for roughly 75 minutes of participation, including a $5 CAD show-up fee.

4.3 Stylized Model

To guide our analysis we present a stylized model where we fit the decision faced by participants of our model to a labour supply model under taxation. Participants choose how many seconds to work t_r out of a maximum of 600 seconds. Subscript $r \in \{1, 3\}$ represents the current round of the participant. Let q_r represent the amount strings

that can be typed in one second.[2] For each string completed the participant earns \$20 dollars and they pay a tax equal to $T_r(20q_r \cdot t_r)$. The exact form of the tax changes in each round. At any point participants may choose to stop but may not resume working after they stop. For every remaining second in the work period the participant earns \$1 dollar. Participants solve the following utility maximization problem:

$$\max_{t_r \in [0,600]} U(t_r) = u(c(t_r)) - \nu(t_r) \quad \text{subject to} \quad c(t_r) = 20q_r \cdot t_r - T_r(20q_r \cdot t_r) + (600 - t_r)$$

where $u(c(t_r))$ is increasing and concave and $\nu(t_r)$ is increasing and convex. The first order condition is

$$u'(c(t_r))[20q_r(1 - T_r'(20q_r \cdot t_r)) - 1] = \nu'(t_r), \tag{4.1}$$

which characterizes the optimal labor supply (working time). Depending on which round the participant is in they faced either no tax, a flat tax, or a progressive tax. We will now discuss how the expression for the optimal labour supply varies under each tax structure.

Case 1 - No Tax:

When no tax is levied on labour income, $T_r(20q_r t_r) = 0$, so the condition characterizing optimal labour supply becomes

$$u'(c(t_r))[20q_r - 1] = \nu'(t_r). \tag{4.2}$$

In this case, the marginal return to work is constant so participants would work until the marginal disutility of labour exceeds the marginal return. In the absence of binding effort costs, participants either work for the entire allotted time or not at all, depending on their productivity q. If we assume that there is no disutility from labour, $\nu(t_r) = 0$, then the optimal labour supply is

$$t^*_{\text{No Tax}} = \begin{cases} 600 & \text{if } 20q_r > 1, \\ 0 & \text{if } 20q_r < 1. \end{cases}$$

Here, as along as a participant is able to complete one string in less than 20 seconds then they would work for the entire period.

[2] We assume this is fixed within each round but may vary between rounds. As we paid 1 lab dollar per second each second participant reallocated from work time to break time, participants should work as fast as they can do to maximize their payoffs no matter how much of labour they intended to supply.

Case 2 - Flat Tax:

When a flat tax is levied than the total tax liability is $T_r^F(20q_r \cdot t_r) = \tau_r \cdot 20q \cdot t_r$ and the expression for the optimal labour supply becomes

$$u'(c(t_r))[20q_r(1-\tau_r)-1] = \nu'(t_r), \tag{4.3}$$

This decision is identical to the case without taxation only the marginal return to working is lowered by the tax τ. If again assume no disutility from labour then the optimal labour supply is given by

$$t^*_{\text{Flat}} = \begin{cases} 600 & \text{if } 20q_r(1-\tau_r) > 1, \\ 0 & \text{if } 20q_r(1-\tau_r) < 1. \end{cases}$$

As our baseline we set $\tau_r = 40\%$. In this setting, a participant should work for the entire period if their speed exceeds one string per 12 seconds.

Case 3 - Progressive Tax:

Under the benchmark progressive tax schedule, the marginal tax rate increases by 10% for every ten strings completed, with a maximum rate of 80% applying from the 81st string onward. The total tax liability is given by

$$T_r^P(20q_r t_r) = \sum_{k=0}^{7}(0.1k)\cdot 20 \cdot b_k(q_r t_r) + 0.8 \cdot 20 \cdot \max\{0, q_r t_r - 80\},$$

where

$$b_k(q_r t_r) = \max\{0, \min(10, q_r t_r - 10k)\}$$

denotes the number of strings completed in tax bracket k.

The condition characterizing optimal labour supply is

$$u'(c(t_r))\left[20q_r(1 - MTR_r^P(q_r t_r)) - 1\right] = \nu'(t_r), \tag{4.4}$$

where the marginal tax rate is given by[3]

$$MTR_r^P(n) = \min\left\{0.1\left\lfloor \frac{n}{10} \right\rfloor, 0.8\right\}.$$

A full derivation of MTR_r^P is provided in Appendix B.

[3]The $\lfloor\ \rfloor$ function outputs the greatest integer less than or equal to the number inside.

Under this tax schedule, participants continue working until the marginal return to an additional unit of effort, net of the $1 stopping payment, falls below the marginal disutility of labour. If we again assume no disutility from working, $\nu(t_r) = 0$, then participants work until

$$20q_r(1 - MTR_r^P(q_r t_r)) \leq 1. \tag{4.5}$$

Under the progressive tax an interior solution would be optimal regardless of the participants disutility of working so long as $q \leq 4$, that is, they cannot complete a string faster than four seconds.

Here we have characterized the optimal labour supply under the benchmark flat tax and progressive tax. These are the tax schedules utilized in round two. In round three, agents would face either a flat or progressive tax calibrated to either the average or marginal tax that they faced on their last completed string in round two. In Section 4.8 we present the optimal labour supply choices under these cases. The same intuition discussed here holds, the differences being that the flat tax levied in round three is not necessarily 40% and the number of bins in the progressive tax schedule within each bracket is not necessarily 10.

4.4 Results

4.4.1 Rationale

This study investigates labour supply responses to changes in tax schedules, focusing on reforms that shift the schedule from linear to nonlinear and vice versa. Given that we individually controlled the marginal piece rate, average wage, and paid 1 lab dollar per second reallocated to break time, labour supply differences were driven by either the changes of average or marginal tax rates under alternative tax schedules, which motivates our design. Our design also allows testing for an alternative behavior model. If participants respond to only one of these tax rates, then treatments that change the linearity of the schedule while varying only the tax rate to which they do not respond should produce similar labour supply choices. We therefore derived our following pre-registered[4] hypotheses (selected) for the cases that people responding only ATR or MTR, or both.

We hypothesized that, relative to labour supply in the previous work shift, or at a given labour supply under the prior tax schedule, participants would complete more

[4] https://osf.io/5s4x9/overview?view_only=80b96a68f2554dfd969d26a38219d1f9

strings when the marginal piece rate for the current string was higher. Similarly, when the average piece rate was higher, participants would also complete more strings. Moreover, consistent with the stylized model, participants were paid 1 lab dollar for each second reallocated to break time and were free to browse the internet or use their phones during the break. We therefore hypothesized that participants would work as quickly as possible, which allows us to use work time — defined as 600 seconds minus the time reallocated to break time — as an alternative measure of labour supply. We present identification results using both measures for comparison.

4.4.2 Summary Statistics

Table 4.1 reports completed strings and work time as measures of labour supply for each work shift by treatment. Across-work-shift variation is greatest in the Prog-Flat-ATR treatment, where participants completed fewer strings and started their breaks earlier under the progressive tax schedule than under the flat tax schedule.

TABLE 4.1: Completed Strings and Work Time by Treatment

	P-F-ATR	P-F-MTR	F-P-ATR	F-P-MTR
Completed Strings				
No Tax	69.44 (26.36)	63.78 (30.04)	63.28 (28.87)	69.89 (24.13)
Round One	62.15 (23.80)	66.03 (20.90)	73.08 (33.36)	82.89 (29.57)
Round Two	86.21 (31.01)	53.70 (41.36)	66.33 (36.04)	82.79 (32.43)
Kruskal–Wallis	19.31	1.23	1.83	7.90
p-value	$< .001$	.54	.40	$<.03$
Work Time				
No Tax	537.62 (150.20)	532.57 (160.44)	506.33 (187.43)	545.29 (120.31)
Round One	433.10 (143.01)	490.52 (129.11)	509.28 (169.33)	547.13 (127.38)
Round Two	533.13 (148.05)	380.32 (263.04)	428.36 (178.46)	519.32 (150.31)
Kruskal–Wallis	22.90	9.49	7.03	1.46
p-value	$< .001$	$< .01$	$< .03$	.48

Notes: Entries report means, with standard deviations in parentheses. In the treatment labels, P denotes a progressive tax schedule, while F denotes a flat tax schedule. Kruskal–Wallis test statistics and corresponding p-values for equality across treatments are reported below the Round Two row for each outcome block.

4.4.3 Preregistered Hypotheses Testing - Pre-reform labour supply reference

Prog-Flat-ATR Treatment

The ATR is controlled, and the MTR is lower than pre-reform. We therefore derive the

following predictions:

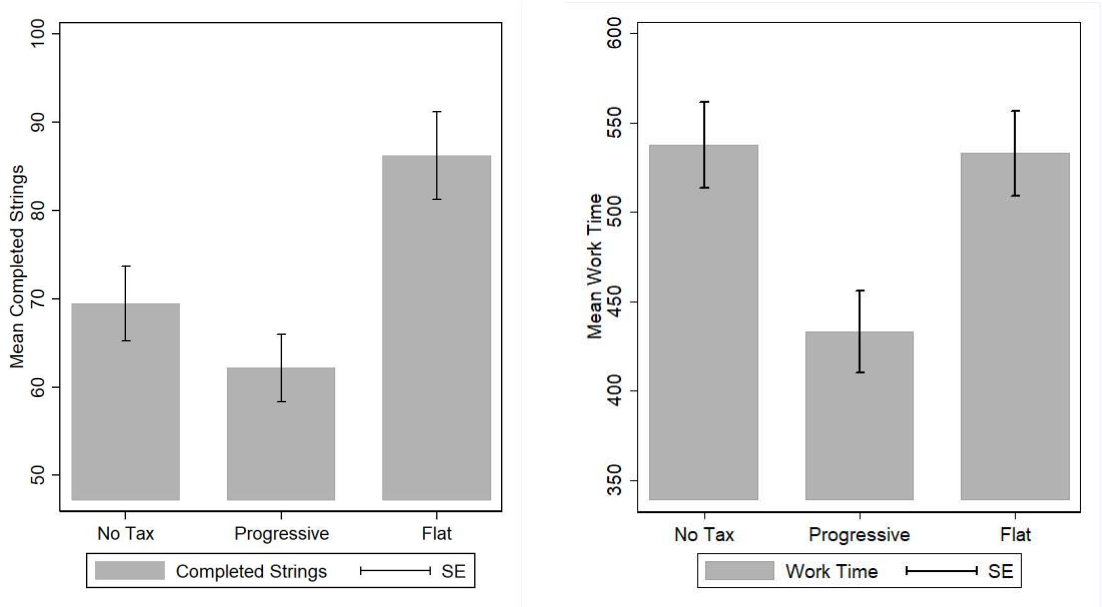
H1 : MTR-focused: $L^2(Progressive) < L^3(Flat)$

H2 : ATR-focused: $L^2(Progressive) = L^3(Flat)$

H1: MTR-focused is verified

Participants, on average, completed 69 strings in round one, 62 strings in round two, and significantly increased to 86 strings in round three (n = 39; Wilcoxon signed-rank: $p < .001$ (positive: 32); Figure 4.2, left), switching from a progressive to an ATR matched flat tax schedule. Similar patterns were shown in participants' work time, from 538 seconds in round one to 433 seconds in round two, and increasing significantly to 533 seconds in round three (n = 39; Wilcoxon signed-rank: $p < .001$ (positive: 29); Figure 4.2, right).

FIGURE 4.2: Labour supplies in Prog-Flat-ATR Treatment



Notes: The histograms show the mean labour supply by rounds in both completed strings (left) and work time (right) when ATR is controlled. The sequences of rounds are no tax, progressive, and flat, respectively.

Flat-Prog-ATR Treatment

The ATR is controlled, and the MTR is higher as long as the number of completed strings is over half the labour supply in round 2. We therefore derive the following predictions:

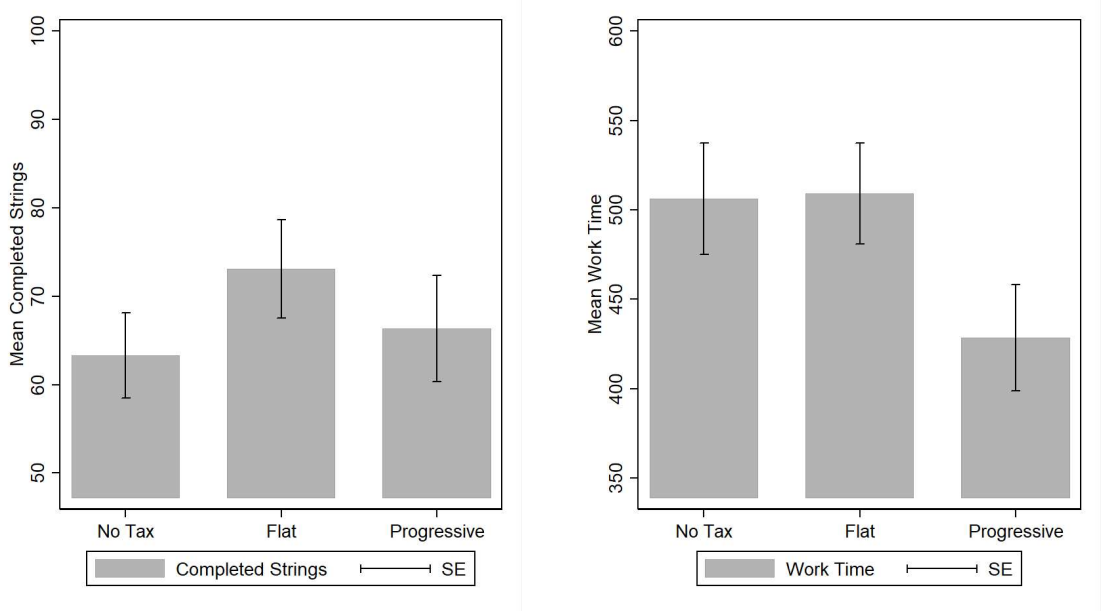
H3 : MTR-focused: $L^2(Flat) > L^3(Progressive)$

H4 : ATR-focused: $L^2(Flat) = L^3(Progressive)$

H3: MTR-focused is verified

Participants, on average, completed 64 strings in round one, 79 strings in round two, and significantly decreased to 68 strings in round three (n = 33 [5]; Wilcoxon signed-rank: $p < .01$ (positive: 9); Figure 4.3, left), switching from a flat to an ATR matched progressive tax schedule. Participants worked 516 seconds in round one, 552 seconds in round two, and their work time increased significantly to 442 seconds in round three (n = 33; Wilcoxon signed-rank: $p < .001$ (positive: 2); Figure 4.3, right).

FIGURE 4.3: Labour supplies in Flat-Prog-ATR Treatment



Notes: The histograms show the mean labour supply by rounds in both completed strings (left) and work time (right) when ATR is controlled. The sequences of rounds are no tax, flat, and progressive, respectively.

Comparing Flat and Progressive when ATR controlled

We developed the ATR-controlled treatments and corresponding hypotheses drawing on the experimental and theoretical approaches of Collins et al. (1992), Sillamaa (1997),

[5]We imposed a minimum bracket size of two strings under the progressive tax schedule in the third round. For the three participants excluded here, we assigned the same progressive tax schedule as in the other two treatments in the second round.

and Carpenter et al. (2016). Result 1, which is more consistent and conclusive, emerges from our individualized within-participant design and from our decision to compensate participants for taking their break early.

Result 1: When the average tax rate (ATR) was held constant, participants significantly reduced labour supply under the progressive tax schedule, indicating responsiveness to changes in the marginal tax rate (MTR).

Prog-Flat-MTR Treatment

The MTR is controlled, and the ATR is higher as long as the number of completed strings is over half the labour supply in round 2. We therefore derive the following predictions:

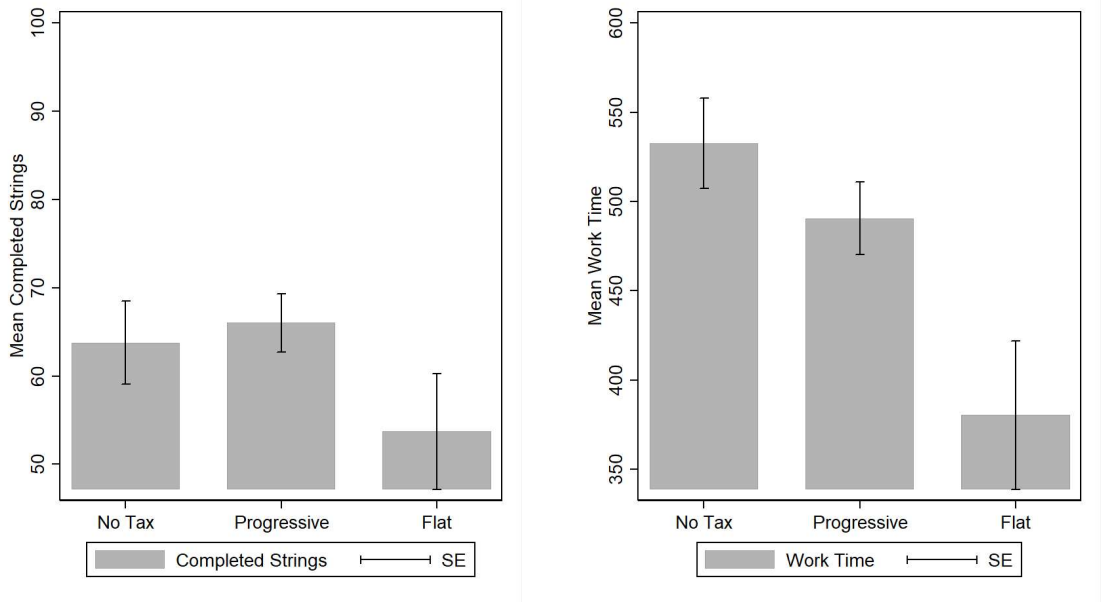
H5 : MTR-focused: $L^2(Progressive) = L^3(Flat)$

H6 : ATR-focused: $L^2(Progressive) > L^3(Flat)$

H5: MTR-focused is verified

Participants, on average, completed 64 strings in round one, 66 strings in round two, and 54 strings in round three (n = 40; Wilcoxon signed-rank: $p = .60$ (positive: 22); Figure 4.4, left), switching from a progressive to an MTR matched flat tax schedule, the increase of ATR did not lead to statistically significant labour response. Similar patterns were shown in participants' work time, from 533 seconds in round one to 491 seconds in round two, and 380 seconds in round three (n = 40; Wilcoxon signed-rank: $p = .21$ (positive: 14); Figure 4.4, right).

FIGURE 4.4: Labour supplies in Prog-Flat-MTR Treatment



Notes: The histograms show the mean labour supply by rounds in both completed strings (left) and work time (right) when MTR is controlled. The sequences of rounds are no tax, progressive, and flat, respectively.

Flat-Prog-MTR Treatment

The MTR is controlled, and the ATR is lower than pre-reform. We therefore derive the following predictions:

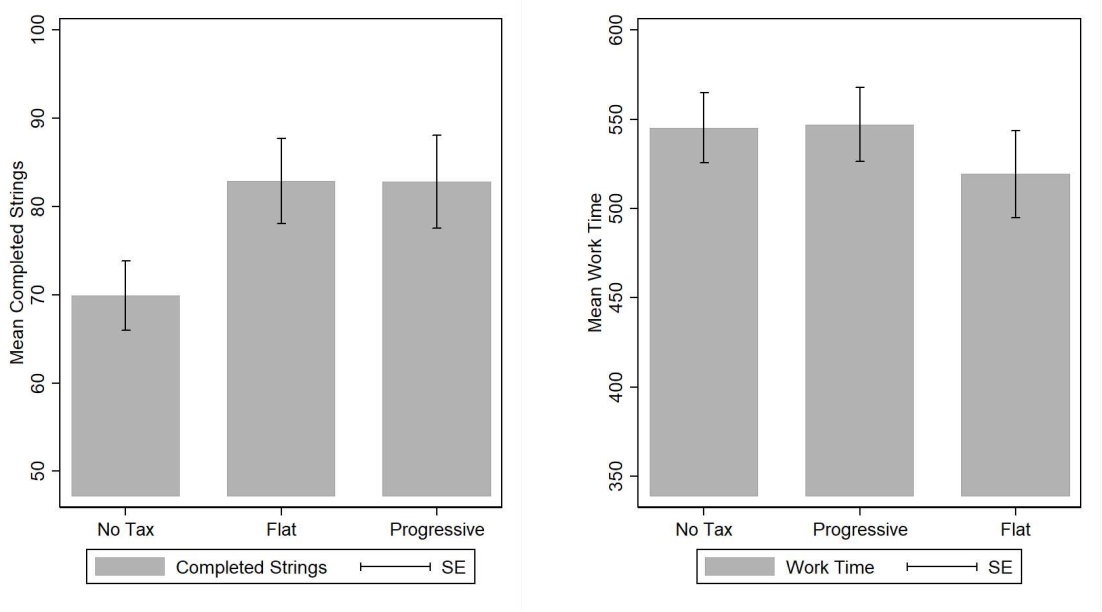
H7 : MTR-focused: $L^2(Flat) = L^3(Progressive)$

H8 : ATR-focused: $L^2(Flat) < L^3(Progressive)$

H7: MTR-focused is verified

Participants, on average, completed 70 strings in round one, 83 strings in round two, and 83 strings in round three (n = 38; Wilcoxon signed-rank: $p = .20$ (positive: 26); Figure 4.5, left), switching from a flat to an MTR matched progressive tax schedule, the decrease of ATR did not lead to statistically significant labour response. In terms of participants' work time, they worked 545 seconds in round one and 547 seconds in round two, and a similar amount of time, 519 seconds, in round three (n = 38; Wilcoxon signed-rank: $p = .14$ (positive: 7); Figure 4.5, right).

FIGURE 4.5: Labour supplies in Flat-Prog-MTR Treatment



Notes: The histograms show the mean labour supply by rounds in both completed strings (left) and work time (right) when MTR is controlled. The sequences of rounds are no tax, flat, and progressive, respectively.

Comparing Flat and Progressive when MTR controlled

We construct the MTR-controlled treatments as the mirror image of the ATR-controlled treatments in order to test whether participants respond to changes in the average tax rate when the marginal tax rate is held constant. These treatments give rise to our second result.

Result 2: When the marginal tax rate (MTR) was held constant, participants did not respond significantly to either changes in progressivity or changes in the average tax rate (ATR).

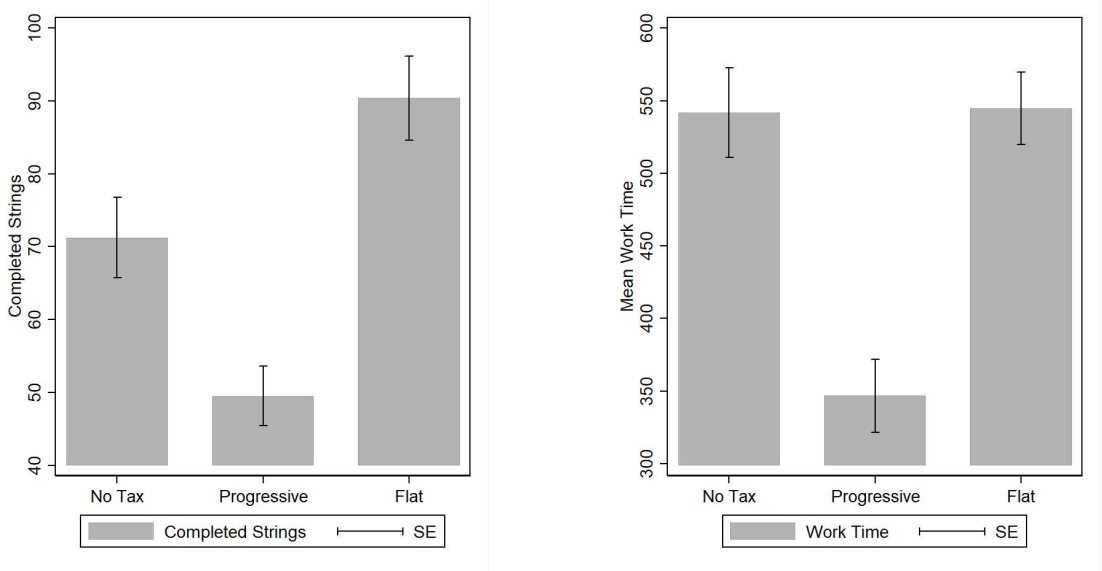
Are all participants MTR-focused? Are there ATR-focused participants?

If a participant is MTR-focused, they should not be willing to work if the MTR is so high that they would earn more by starting the break. Paying one lab dollar per second for the remaining work time allows us to identify differences in labour supply responses while identifying participants who are sensitive to the cost of labour. For example, the highest marginal tax rate in our design is 80 percent, under which participants earn

four lab dollars per completed string. Across all 153 participants in our treatments, no participant achieved an average completion speed faster than four seconds per string in any ten-minute work period. We could therefore classify participants who continue working under an 80 percent tax rate as participants insensitive to the marginal piece rate, i.e. piece rate per completed string minus MTR, compared to the labour cost. We extended this categorization by calculating participants' typing speed under the progressive tax schedule that reached an 80% tax rate, and compared their speed to the marginal tax rate they faced. If they stopped before or at the point where what they could earn by completing strings was higher than or equal to taking the break, we define them as participants who are indeed MTR-focused (20 lab dollars net the marginal tax rate), and we examine whether the groups respond differently to tax schedule changes. We begin by repeating the tests above with our more restricted MTR-focused group.

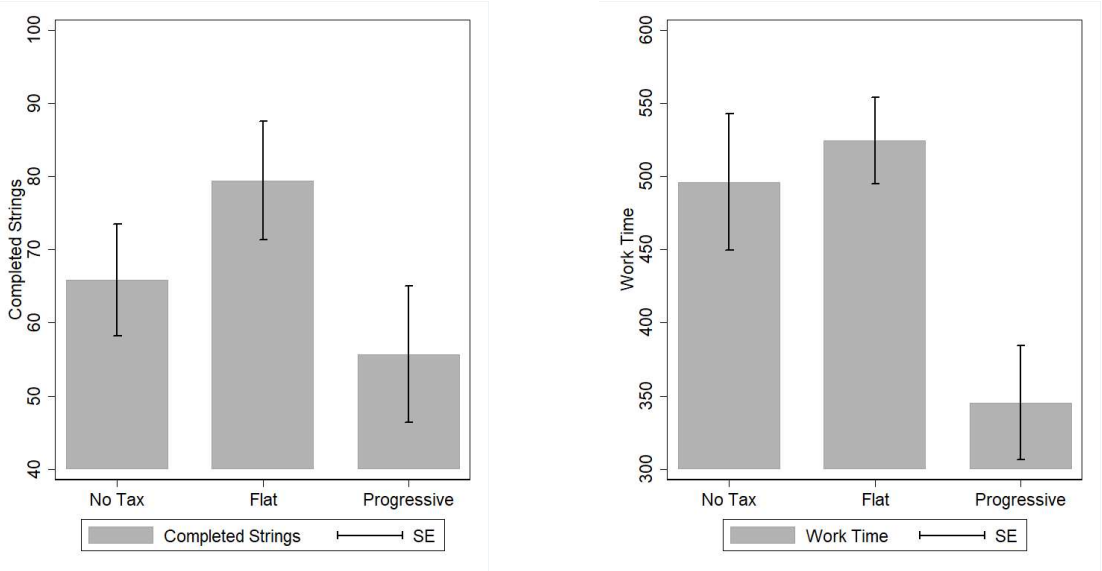
In **Prog-Flat-ATR**, the mean number of completed strings increased significantly when the tax schedule changed from progressive to flat (n = 23 (MTR-focused), from 50 to 90, Wilcoxon signed-rank: $p < .001$ (positive: 20); Figure 4.6 left). Similarly, mean work time increased significantly (from 347 to 545, Wilcoxon signed-rank: $p < .001$ (positive: 21); Figure 4.6 right). The pattern is robust in reverse order in **Flat-Prog-ATR** (n = 17 (MTR-focused), from 79 to 56, Wilcoxon signed-rank: $p < .001$ (negative: 17), Figure 4.7 left; from 525 to 345, Wilcoxon signed-rank: $p < .001$ (negative: 16), Figure 4.7 right).

FIGURE 4.6: Labour supplies in Prog-Flat-ATR Treatment (MTR-focused)



Notes: The histograms show the mean labour supply of MTR-focused participants by rounds in both completed strings (left) and work time (right) when ATR is controlled. The sequences of rounds are no tax, progressive, and flat, respectively.

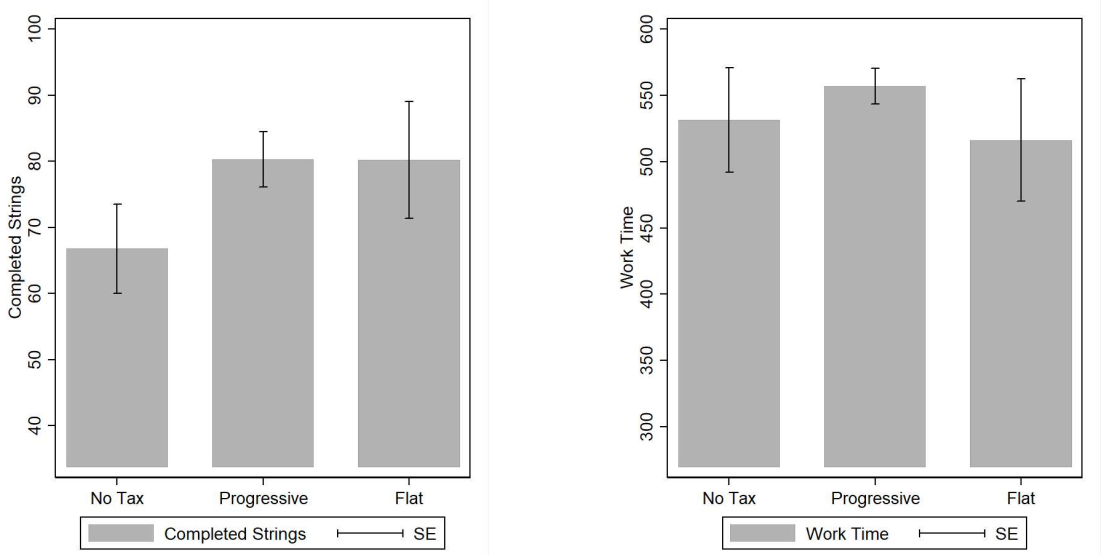
FIGURE 4.7: Labour supplies in Flat-Prog-ATR Treatment (MTR-focused)



Notes: The sequences of rounds are no tax, flat, and progressive, respectively.

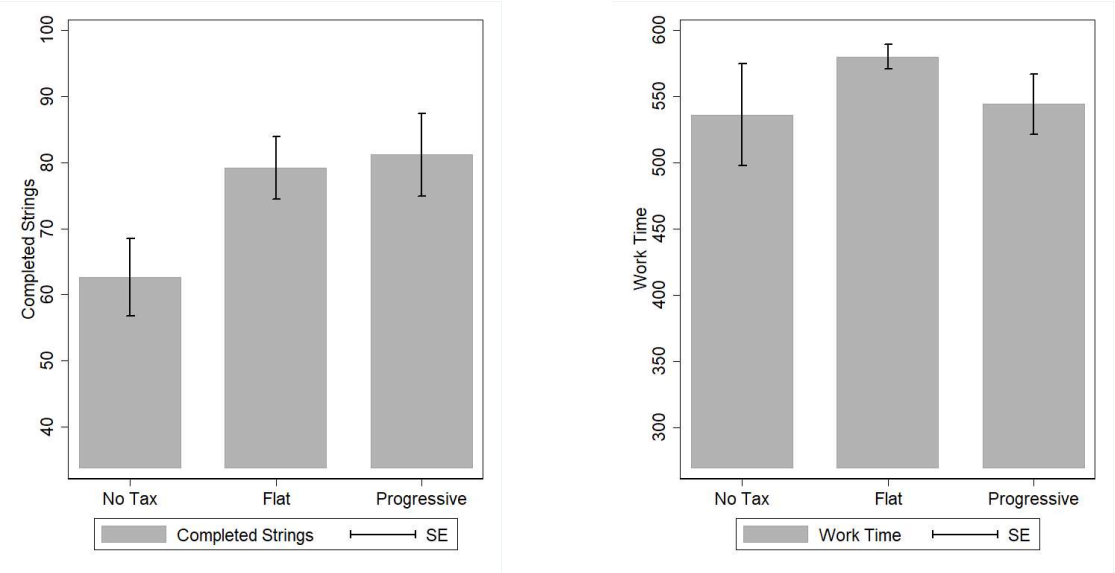
However, among participants who are not in our more restricted MTR-focused group. In **Prog-Flat-ATR**, the mean number of completed strings was similar under the progressive and flat tax schedules (n = 16, from 80 to 80, Wilcoxon signed-rank: $p = .30$ (positive: 12); Figure 4.8 left). Similarly, the mean work time is also not statistically significant (from 557 to 516, Wilcoxon signed-rank: $p = .41$ (positive: 8); Figure 4.8 Right). Similar patterns were found with the reverse order in **Flat-Prog-ATR** (n=16, from 79 to 81, Wilcoxon signed-rank: $p = .59$ (positive: 9); Figure 4.9 left; from 580 to 544, Wilcoxon signed-rank: $p = .16$ (positive: 2); Figure 4.9 right).

FIGURE 4.8: Labour supplies in Prog-Flat-ATR Treatment (non-MTR-focused)



Notes:The histograms show the mean labour supply of non-MTR-focused participants by rounds in both completed strings (left) and work time (right) when ATR is controlled. The sequences of rounds are no tax, progressive, and flat, respectively.

FIGURE 4.9: Labour supplies in Flat-Prog-ATR Treatment (non-MTR-focused)



Notes: The sequences of rounds are no tax, flat, and progressive, respectively.

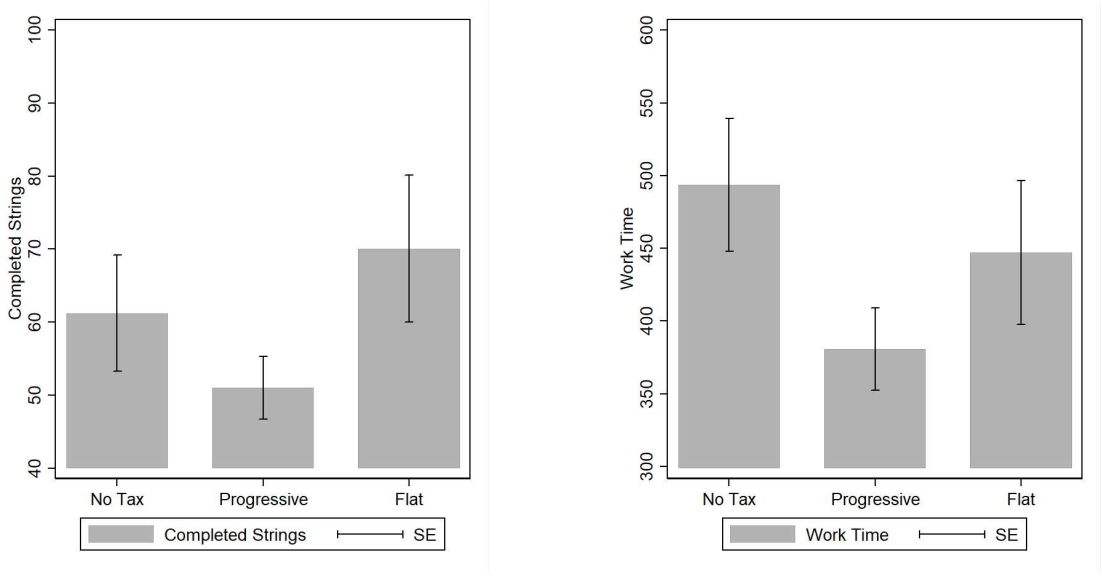
Comparing MTR-focused with non-focused group when ATR is controlled
After recategorizing participants into MTR-focused and non-MTR-focused groups based on their labour supply under the progressive tax schedule, we find that Result one is driven by the MTR-focused group.

Result 3: Participants who are sensitive to the trade-off between the marginal piece rate and the cost of labour (the MTR-focused group) increased labour supply under the flat tax schedule relative to the progressive schedule, whereas the non-MTR-focused group did not respond to MTR changes.

On the other hand, for the MTR-focused group in **Prog-Flat-MTR**, the mean number of completed strings marginally increased from the progressive to flat tax schedules (n = 16, from 51 to 70, Wilcoxon signed-rank: $p < .05$ (positive: 13); Figure 4.10 left), but the difference in mean work time is not statistically significant (from 381 to 447, Wilcoxon signed-rank: $p = .08$ (positive: 12); Figure 4.10 right). On the contrary, for the non-MTR-focused group in **Prog-Flat-MTR**, the mean number of completed strings decreased significantly when the tax schedule changed from progressive to flat (n =23, from 77 to 42, Wilcoxon signed-rank: $p < .03$ (positive: 9); Figure 4.11 left). Mean work

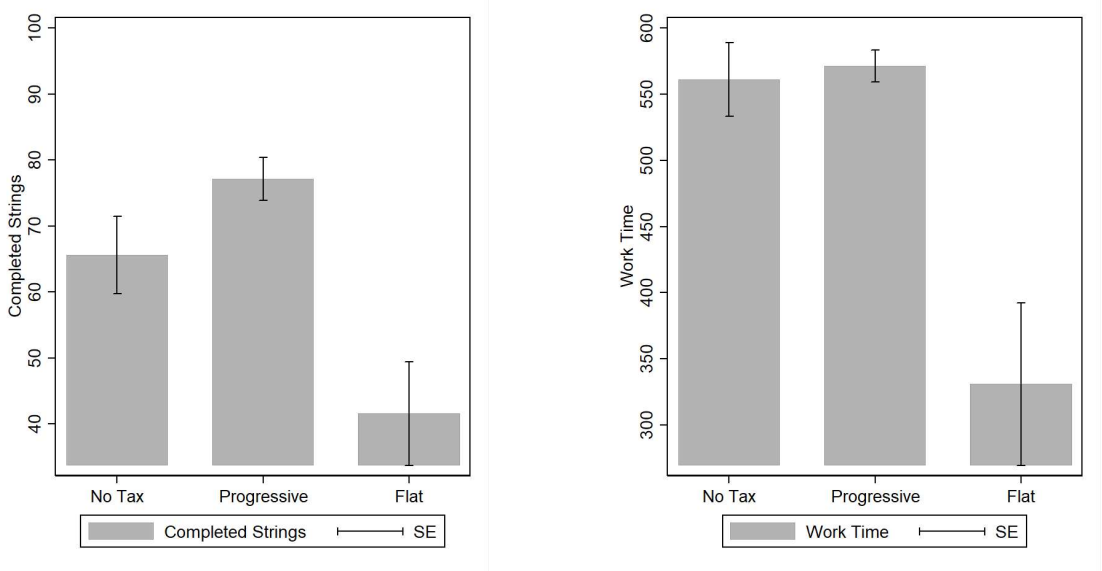
time significantly decreased (from 572 to 331, Wilcoxon signed-rank: $p < .01$ (positive: 2); Figure 4.11 right).

FIGURE 4.10: Labour supplies in Prog-Flat-MTR Treatment (MTR-focused)



Notes: The histograms show the mean labour supply of MTR-focused participants by rounds in both completed strings (left) and work time (right) when MTR is controlled. The sequences of rounds are no tax, progressive, and flat, respectively.

FIGURE 4.11: Labour supplies in Prog-Flat-MTR Treatment (non-MTR-focused)



Notes: The histograms show the mean labour supply of non-MTR-focused participants by rounds in both completed strings (left) and work time (right) when MTR is controlled. The sequences of rounds are no tax, progressive, and flat, respectively.

Comparing MTR-focused with non-focused group across treatments

One might argue that Result three is endogenous and partly mechanical, given how the groups are defined. Even so, inference remains possible if we compare the same subgroup across treatments. Notably, MTR-focused participants respond to changes in the marginal tax rate (MTR), but not to changes in the average tax rate (ATR).[6] By contrast, the non-MTR-focused group responds to changes in the ATR. Under our framework, these participants are effectively ATR-focused: they supply significantly more labour when the average tax rate is low and reduce labour supply when the average tax rate is high, while remaining insensitive to the trade-off between the marginal piece rate and the cost of labour under the progressive tax schedule.[7]

Result 4: MTR-focused participants are not responsive to ATR changes,

[6]This result is sensitive to the identification strategy, but the direction is nonetheless opposite to that of the non-MTR-focused group.

[7]By design, we do not observe participants working more slowly than the payoff-implied cutoff in the highest marginal tax rate bracket of the **Flat-Prog-MTR** treatment; as a result, our categorization has no identifying power in this treatment.

whereas non-MTR-focused participants are effectively ATR-focused, responding to ATR changes but not to MTR changes.

4.4.4 Heterogeneous labour responds to change of taxes

We hypothesized that individuals may respond to changes in the ATR, the MTR, or both, relative to their pre-reform labour supply. We find heterogeneous effects based on participants' sensitivity to the marginal piece rate compared to the labour cost, as induced in the laboratory environment. A natural question, then, is how these findings map onto the stylized model. In this section, we utilize the model presented in section 4.3 to calculate the optimal working time and strings completed for our participants. To simplify, we assume there is no effort cost to complete the strings other than the opportunity cost of time; we assume participants type as fast as they can, and that utility is linear in consumption. In section 4.3, we assume that under a flat tax, participants should work for the full session if

$$20q_r(1-\tau_r) > 1,$$

The agent works fast enough such that the marginal return from completing an additional string exceeds the \$1 dollar-per second they could receive from not working. For each round in which a participant faced a flat tax schedule, they based their decision on their working speed to determine whether they should have worked for the entire work session, given the tax rate that they faced. On the other hand, under the progressive tax, participants should continue working as long as

$$1 - \frac{1}{20q} > MTR_r^P(qt_r),$$

that is, as long as the marginal tax rate remains below the threshold at which the after-tax return to an additional unit of work falls below the \$1 stopping payment. For rounds where participants face progressive tax schedules, they based their decision on their work speed to calculate the marginal tax rate under which they should stop working.

The conditions here simply imply that participants should decide whether to continue working or begin their break based on the piece rate for the current string and their

typing speed. Alternatively, based on our behavioral rationale and findings, we may instead formulate the conditions as follows:

$$Keep\ working\ if \begin{cases} 20q_r(1-\tau_r) > 1, & \text{if ATR-focused} \\ 1 - \frac{1}{20q} > MTR_r^P(qt_r), & \text{if MTR-focused} \end{cases}$$

We can then map the heterogeneous labour supply responses of the two groups onto the stylized model by interpreting marginal-incentive-sensitive participants as MTR-focused and non-MTR-focused participants as ATR-focused. MTR-focused participants decide whether to continue working based on the marginal tax rate in the current bracket and their typing speed. When the tax schedule changes from progressive to flat, holding the ATR constant implies that these participants face a lower MTR at the point where they previously stopped working, while the ATR remains unchanged. The lower MTR may then induce them to continue working throughout the shift. By contrast, when the MTR is matched, participants should continue working throughout the shift. The same logic applies in the reverse order.

By contrast, ATR-focused participants decide whether to continue working based on the current average tax rate and their typing speed. Consider the **Prog-Flat-ATR** for example, 9 of 40 participants completed more than 80 strings under the progressive tax schedule, thereby working in the range associated with an 80 percent MTR. Their average completion times ranged from 5.00 to 7.41 seconds per string. We classify these participants as non-MTR-focused because, at an 80 percent tax rate, they earn only four lab dollars per completed string and would have been better off stopping earlier. The ATR they faced ranged from 36 percent to 50 percent. Moving to a 40 percent flat tax rate, under which they earn 12 lab dollars per string, does not substantially change their labour supply, even though the MTR falls sharply from 80 percent to 40 percent. This is consistent with insensitivity to MTR in Round 2 under the progressive tax schedule. The same reasoning also explains why the number of completed strings decreases sharply in Round 3 of the **Prog-Flat-MTR** treatment, where these participants would face an 80 percent flat average tax rate. The same logic applies in the reverse order.

Taken together, these results suggest that, in our experiment, MTR-focused participants respond more strongly to changes in the linearity of the tax schedule than to changes in the overall tax burden. By contrast, ATR-focused participants respond more strongly to changes in the overall tax burden than to the distinction between progressive and flat tax schedules.

4.5 Discussion and Conclusion

Despite the richness of economic theory and the large literature on labour supply responses to taxation, we know comparatively little about how individuals respond when tax schedules shift between linear and nonlinear forms, especially when they face little friction in adjusting labour supply. Admittedly, the multiple theoretical mechanisms that operate simultaneously under such schedule changes make clean isolation difficult. This motivates our use of behavioral control and experimental design rather than an attempt to isolate each mechanism separately. In this paper, we hold either the average tax rate (ATR) or the marginal tax rate (MTR) constant within participants in order to identify differences in labour supply responses to changes in the linearity of the tax regime.

Our first and second result is broadly consistent with the literature, suggesting that individuals are generally more responsive to MTR than to ATR. At the same time, responses to ATR appear to be more sensitive to the surrounding institutional setting. One feature that distinguishes our design from most related laboratory studies is that we compensate participants for time allocated to rest, thereby inducing an explicit value of leisure (Blumkin et al., 2012). Paying participants one lab dollar per second of rest allows us to identify a clear threshold beyond which continued work is no longer privately profitable.

One question of particular interest is whether a participant who stops working at a given MTR would also stop working if later faced with that same MTR at the point corresponding to their highest prior level of labour supply. Our third and fourth results show that our initial conjecture was too strong: not all participants stop working once continued work is no longer privately profitable. Among MTR-focused participants, labour supply is similar when MTR is matched. By contrast, ATR-focused participants do not stop at the first point at which continued work becomes unprofitable. Instead, they continue working until, in the following round, they face an ATR that, given the matched MTR, is sufficiently high to induce stopping. The fact that these participants continue working even when doing so is no longer privately profitable, yet stop under sufficiently high ATR, helps rule out simple inattention to the payment scheme or the possibility that they were merely working because they enjoyed the task.

Our heterogeneous results may help reconcile some of the inconsistent findings in the existing literature. Even in a laboratory setting in which adjustment frictions are largely removed, taxes do not finance any public good or other use, and the task is

unfamiliar to most participants, we still observe substantial heterogeneity in labour supply responses to taxation. Further work is needed to assess the robustness of these findings across settings, but we hope this study encourages future research to consider ex ante or ex post stratification by behavioral type. More ambitiously, our findings may also inform public finance policy when policymakers consider reforms that alter the structure of the tax system from linear to nonlinear, or vice versa, and evaluate the associated welfare consequences across different behavioral groups. In our setting, MTR-focused participants are sensitive to the linearity of the tax schedule but relatively insensitive to the overall tax burden, whereas ATR-focused participants remain willing to work under progressive taxation but are sensitive to the overall tax burden. If these groups can be identified empirically, and if our results prove robust, the distinction may be relevant for policy design.

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4.6 Instructions

4.6.1 Instruction – Main

Instruction – General:

Welcome, and thank you for participating in this decision-making experiment. Please make all decisions on your own and do not consult or discuss the experiment with anyone else.

At the end of today's session, you will receive a $5 participation payment, plus any additional earnings determined by the decisions you make and certain random factors. Please read the instructions carefully. A good understanding of the instructions and thoughtful decisions may increase the amount of cash you receive in addition to your participation payment.

All responses and information you provide in this study will be kept anonymous and used for research purposes only.

Today's experiment consists of three work shifts and a questionnaire. You will earn money based on your productivity in the task and receive a $5 participation payment. Payment for the task is subject to tax, which may vary across work shifts. The payment and tax system for each work shift will be explained before that work shift begins.

You will have the opportunity to work on a task for up to 30 minutes in total, three 10-minute work shifts. After the first 10-minute work shift, you will have a 5-minute break. After the second 10-minute work shift, you will have another 5-minute break. The third 10-minute work shift will follow the second break.

However, you may choose to end any work shift early and move to the break before the 10 minutes are over. Any remaining time from that work shift — including the third work shift — will be added to your break time. If there is no remaining time in the third work shift, there will be no third break. This means that the experiment will not end earlier if you stop working before the shift ends. Instead, taking a break before the shift ends simply changes your time allocation between work and break. During the break, you may use your cellphone or browse the internet.

At the end of the experiment, you will learn the outcomes of your decisions and be informed of your final payment.

If you have any questions about the instructions, please raise your hand and someone will assist you.

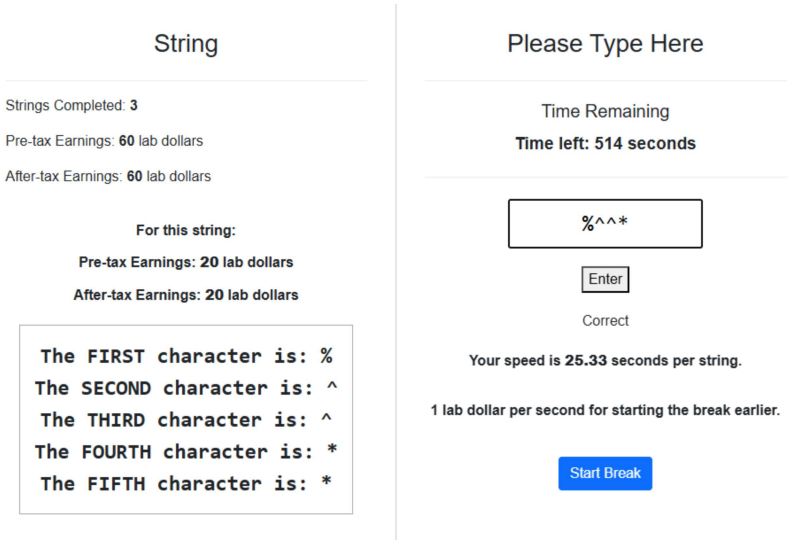
Task:

The task involves typing a string of five special characters into a text box. These special

characters are the symbols that appear when you press the number keys from 1 to 9 while holding the Shift key.
When you have finished typing the character string, either click the Enter button on the screen or press the Enter key on your keyboard.
Each time you correctly type all five characters in the correct order and press Enter, a new string of five characters will appear.
On the left side of the task interface, the five characters are displayed vertically, with the first character at the top and the fifth character at the bottom. Above the characters, you will also see:

1. the number of strings you have completed during the shift,
2. your pre-tax earnings,
3. your after-tax earnings for that part, and
4. how much you can earn, before tax and after tax, for completing the current string.

On the right side of the task interface, you will see the text entry box and the Enter button. Above them, a countdown timer will show how much time remains in the work shift. Below the entry box, you will see your average speed for completing a string, as well as the Start Break button. **Screenshot of the task interface:**



Earnings – No Tax:
We will use lab dollars as the unit of payment during the experiment. One lab dollar =

$0.005 CAD (Half cent).
For each string you complete during the first 10-minute work shift, you will earn 20 lab dollars.
If, at any point during the 10-minute work shift, you would like to stop working on the task, you may press the Start Break button and use the internet. Any time remaining in your work shift when you press the Start Break button will be added to your 5-minute break.
In addition, each second remaining in your work shift when you press the Start Break button is worth 1 lab dollar.

Your earnings from this work shift are calculated as follows:
Number of completed strings × 20 + seconds remaining when you press "Start Break" = Your Earnings

To switch from the experiment tab to a standard web browser, press Alt + Tab at the same time. You can switch back by pressing Alt + Tab again.
If you have a question at any time, please raise your hand, and an experimenter will come over to help you individually.

4.6.2 Instruction – Flat

Earnings – Flat Tax:
For each string you complete during the next 10-minute work shift, you will earn 20 lab dollars before tax. 50 percents of this amount will be taxed by the experimenter, so your after-tax earnings will be 10 above lab dollars for each completed string.
As in the previous work shift, if at any point during the 10-minute work shift you would like to stop working on the task, you may press the Start Break button and use the internet. Any time remaining in your work shift when you press the Start Break button will be added to your next 5-minute break.
In addition, each second remaining in your work shift when you press the Start Break button is worth 1 lab dollar. This amount will not be taxed.

Your earnings from this work shift are calculated as follows:
Number of completed strings × 10 + seconds remaining when you press "Start Break" = Your Earnings

To switch from the experiment tab to a standard web browser, press Alt + Tab. To switch back, press Alt + Tab again.
If you have a question at any time, please raise your hand, and an experimenter will assist you.

4.6.3 Instruction – Progressive

Earnings – Increasing Tax:
For each string you complete during the next 10-minute work shift, you will earn 20 lab dollars before tax. The tax charged by the experimenter increases as the number of completed strings increases. Therefore, your after-tax earnings per string will decrease across ranges, from 20 lab dollars to 4 lab dollars.
After-Tax Earnings Schedule

- **Strings 1–10:** 20 lab dollars each
- **Strings 11–20:** 18 lab dollars each
- **Strings 21–30:** 16 lab dollars each
- **Strings 31–40:** 14 lab dollars each
- **Strings 41–50:** 12 lab dollars each
- **Strings 51–60:** 10 lab dollars each
- **Strings 61–70:** 8 lab dollars each
- **Strings 71–80:** 6 lab dollars each
- **Strings 81 and above:** 4 lab dollars each

The tax applies only to strings within the relevant range. For example, you will still earn 20 lab dollars for each of your first 10 completed strings, regardless of how many strings you complete in total during this shift. Similarly, you will earn 18 lab dollars for the next 10 strings (i.e., strings 11, 12, 13, ... , 20), even if you complete more than 20 strings in total.
As in the previous work shift, if at any point during the 10-minute work shift you would like to stop working on the task, you may press the Start Break button and use the internet. Any time remaining in your work shift when you press the Start Break button will be added to your next 5-minute break.
In addition, each second remaining in your work shift when you press the Start Break button is worth 1 lab dollar. This amount will not be taxed.

Your earnings from this work shift are calculated as follows:
(Strings 1–10 × 20) + (Strings 11–20 × 18) + (Strings 21–30 × 16) + ... +

seconds remaining when you press "Start Break" = Your Earnings

To switch from the experiment tab to a standard web browser, press Alt + Tab at the same time. You can switch back by pressing Alt + Tab again.
If you have a question at any time, please raise your hand, and an experimenter will come over to help you individually.

4.7 Progressive Tax MTR Calculation

For the benchmark progressive tax schedule, to derive the marginal tax rate start by taking the derivative of the step-function $b_k(q_r \cdot t_r)$. let $n = q_r \cdot t_r$ represent the number of strings completed, the step function is

$$b_k(n) = \begin{cases} 0 & n \leq 10k \\ n - 10k & 10k < n < 10(k+1) \\ 10 & n \geq 10(k+1). \end{cases}$$

Because the tax schedule is a step function, the derivative is not defined at bracket thresholds. We therefore use the right-hand derivative, which captures the tax applied to the next string completed. The right-hand derivative of $b_k(n)$ with respect to number of strings completed is

$$b_k^+(n) = \begin{cases} 0 & n < 10k \\ 1 & 10k \leq n < 10(k+1) \\ 0 & n \geq 10(k+1). \end{cases}$$

Now taking the right-hand derivative of the progressive tax function with respect to the number of strings yields

$$\frac{d^+ T_r^P(20n)}{dn} = \sum_{k=0}^{7} (0.1k) \cdot 20 \cdot b_k^+(n) + 0.8 \cdot 20 \cdot \mathbb{1}\{n \geq 80\}.$$

Dividing this function by 20 gives us the marginal tax rate per additional dollar earned,

$$MTR_r^P(n) = \sum_{k=0}^{7} (0.1k) \cdot b_k^+(n) + 0.8 \cdot \mathbb{1}\{n \geq 80\}.$$

Notice that

$$\sum_{k=0}^{7}(0.1k)\cdot b_k^+(n)=\sum_{k=0}^{7}(0.1k)\cdot \mathbb{1}\{10k\leq n<10(k+1)\},$$

for any $n<80$, each k corresponds to a unique bracket $10k\leq n<10(k+1)$, so exactly one $b_k^+(n)$ equals one and the entire sum becomes $0.1k$. Isolating k in $10k\leq n<10(k+1)$ yields $k\leq \frac{n}{10}<k+1$ which can be represented by $\left\lfloor\frac{n}{10}\right\rfloor$. Also, for any $n\geq 80$ the marginal tax rate is 0.8. Given these two facts, the marginal tax rate for the progressive tax system is equal to

$$MTR_r^P(n)=\min\{0.1\left\lfloor\frac{n}{10}\right\rfloor,0.8\}.$$

4.8 Alternative Tax Calibrations

Here we present the analytical expressions that characterize the optimal labour supply under round three tax schedules that are designed to match either the average tax rate or the marginal tax rate experienced by the participant in round two.

4.8.1 Progressive-Flat-ATR

In the Progressive-Flat-ATR treatment participants first complete the real effort task while there is no tax levied. In round two they complete the same task under the benchmark progressive tax schedule. In round three the tax levied on their labour income is a flat tax set to be equal to the average tax rate on their labour income from round two. Formally,

$$\tau_3=\frac{T_2^P(20q\cdot t_2^*)}{20q\cdot t_2^*},$$

where t_2^* is the amount of time spent working in round two. The expression for optimal labour supply is the same as Equation (4.3) and the decision rule remains the same as well. Participants should work up until the point where the marginal cost of working exceeds the marginal return from working. If we assume there is no cost to working, then participants should work entirely or not at all depending on whether the tax rate results on their marginal return from working is above or below \$1 dollar, respectively. However, because the flat tax in this treatment is equal to the average tax rate from the previous round we know that

$$\tau_3<MTR_2^P,$$

the tax rate faced in round three is less than the marginal tax rate faced by the participant on their last string completed. This suggests that if there is no cost to working

then participants should always choose to work the entire work period in the third round conditional on at least reaching the first tax bracket in round two.

4.8.2 Progressive-Flat-MTR

In the Progressive-Flat-MTR treatment participants first complete the real effort task while there is no tax levied. In round two they complete the same task under the benchmark progressive tax schedule. In round three the tax levied on their labour income is a flat tax set to be equal to the marginal tax rate on their labour income from round two. Formally,

$$\tau_3 = \frac{\partial T_2^P(20q \cdot t_2^*)}{\partial(20q \cdot t_2^*)},$$

where t_2^* is the amount of time spent working in round two. Here we can see that τ_3 would be set to equal to the tax rate levied on the last string completed by the participant in round two. The expression for optimal labour supply is the same as Equation (4.3) and the decision rule remains the same as well. Participants should work up until the point where the marginal cost of working exceeds the marginal return from working. If we assume there is no cost to working, then participants should work entirely or not at all depending on whether the tax rate results on their marginal return from working is above or below \$1 dollar, respectively. However, the flat tax in this treatment is equal to the marginal tax rate faced by the participant on their last string completed in round. This means that the tax rate in three is constant at a level that the participant was previously willing to work. This suggests that if there is no cost to working then participants should always choose to work the entire work period in the third round conditional on working at all in round two.

4.8.3 Flat-Progressive-ATR

In the Flat-Progressive-ATR treatment participants first complete the real effort task while there is no tax levied. In round two they complete the same task under the benchmark flat tax. In round three the tax levied on their labour income is a progressive tax schedule designed so that if the participant completes the same amount of strings in round three that they did in round two then the average tax rate on their labour earnings in round three would be the same as it was in round two. Formally,

$$\frac{T_3^P(20q_2 \cdot t_2^*)}{20q_2 \cdot t_2^*} = \frac{T_2^F(20q_2 \cdot t_2^*)}{20q_2 \cdot t_2^*} = \tau_2.$$

To achieve this we alter bin size of the tax brackets, keeping the step size at 10% per bin as well as keeping nine total tax brackets. Let $n_2 = q_2 \cdot t_2^*$ be the number of strings completed by a participant in round two. For each participant we first calculate

$$a = \left\lfloor \frac{n_2 - 1}{8} \right\rfloor.$$

and

$$s = (n_2 - 1) \bmod 8$$

to determine the minimum size of each bin and the remainder, respectively. The bin size w_k is then equal to

$$w_k = a + \mathbb{1}\{k \leq s\}, \quad k = 1, ..., 8$$

and $w_9 = \infty$. Each bracket gets the same baseline size a, and the first s brackets get one extra unit. The first eight brackets have finite width, while the ninth bracket is open-ended. Define

$$U_k = \sum_{i=1}^{k} w_i, \quad k = 1, ..., 8,$$

with $U_0 = 0$ as the cumulative size of all the brackets excluding the last one. The marginal tax rate in each bin is then

$$\tau_j = \begin{cases} 0 & j = 1 \\ \frac{j-1}{10} & j = 2, ..., 9. \end{cases}$$

Finally, this means that the total tax liability is equal to

$$T_3^P(20qt_3) = \sum_{k=1}^{8} \tau_k \cdot 20 \cdot \max\{0, \min(w_k, q_3 \cdot t_3 - U_{k-1})\} + 0.8 \cdot 20 \cdot \max\{0, q_3 \cdot t_3 - U_8\}.$$

Next, we want to derive an expression for the marginal tax rate in this setting. let $n_3 = q_3 \cdot t_3$ represent the number of strings completed in round three. The taxable amount in bracket k is

$$b_k(n) = \max\{0, \min(w_k, n_3 - U_{k-1})\}, \qquad k = 1, \ldots, 8.$$

Equivalently,

$$b_k(n_3) = \begin{cases} 0 & n_3 \leq U_{k-1}, \\ n_3 - U_{k-1} & U_{k-1} < n_3 < U_k, \\ w_k & n_3 \geq U_k. \end{cases}$$

Because the tax schedule is a step function, the derivative is not defined at bracket thresholds. We therefore use the right-hand derivative, which captures the tax applied to the next string completed. The right-hand derivative of $b_k(n_3)$ with respect to the number of strings completed is

$$b_k^+(n) = \begin{cases} 0 & n_3 < U_{k-1}, \\ 1 & U_{k-1} \leq n_3 < U_k, \\ 0 & n_3 \geq U_k. \end{cases}$$

Now taking the right-hand derivative of the progressive tax function with respect to the number of strings yields

$$\frac{d^+ T_3^P(20n_3)}{dn_3} = \sum_{k=1}^{8} \tau_k \cdot 20 \cdot b_k^+(n_3) + 0.8 \cdot 20 \cdot \mathbb{1}\{n_3 \geq U_8\}.$$

Dividing by 20 gives the marginal tax rate per additional dollar earned:

$$MTR_3^P(n_3) = \sum_{k=1}^{8} \tau_k b_k^+(n_3) + 0.8 \cdot \mathbb{1}\{n_3 \geq U_8\}.$$

Since $\tau_k = (k-1)/10$, this can be written as

$$MTR_3^P(n_3) = \sum_{k=1}^{8} \frac{k-1}{10} \mathbb{1}\{U_{k-1} \leq n_3 < U_k\} + 0.8 \cdot \mathbb{1}\{n_3 \geq U_8\}.$$

For any $n_3 < U_8$, exactly one bracket indicator equals one. Therefore the marginal tax rate is the marginal tax rate associated with the bracket containing n. For any $n_3 \geq U_8$, the marginal tax rate is 0.8. Hence,

$$MTR_3^P(n_3) = \begin{cases} \frac{k-1}{10} & \text{if } U_{k-1} \leq n_3 < U_k, \quad k = 1, \ldots, 8, \\ 0.8 & \text{if } n_3 \geq U_8. \end{cases}$$

Finally, the condition characterizing the optimal labour supply in this setting is

$$u'(c(t_3))\left[20q_3(1 - MTR_3^P(q_3t_3)) - 1\right] = \nu'(t_3), \tag{4.6}$$

4.8.4 Flat-Progressive-MTR

In the Flat-Progressive-MTR treatment participants first complete the real effort task while there is no tax levied. In round two they complete the same task under the benchmark flat tax. In round three the tax levied on their labour income is a progressive tax schedule designed so that if the participant completes the same amount of strings in round three that they did in round two then the marginal tax rate on their labour earnings in round three would be the same as it was in round two. Formally,

$$MTR_3^P(q_2 \cdot t_2^*) = MTR_2^F(q_2 \cdot t_2^*) = \tau_2,$$

To achieve this we alter bin size of the tax brackets, keeping the step size at 10% per bin as well as keeping nine total tax brackets. Let $n_2 = q_2 \cdot t_2^*$ be the number of strings completed by a participant in round two. For each participant we first calculate

$$a = \left\lfloor \frac{n_2 - 1}{4} \right\rfloor,$$

and

$$s = (n_2 - 1) \bmod 4$$

to determine the minimum size of each bin and the remainder, respectively. The bin size w_k are defined symmetrically so that

$$w_k = a + \mathbb{1}\{k \leq s\}, \quad k = 1, ..., 4$$

and

$$w_k = w_{k-4}, \quad k = 5, ..., 8,$$

with $w_9 = \infty$. Each bracket gets the same baseline size a, and the first s lower brackets, and their symmetric counterparts, get one extra unit. The first eight brackets have finite width, while the ninth bracket is open-ended. Define

$$U_k = \sum_{i=1}^{k} w_i, \quad k = 1, ..., 8,$$

with $U_0 = 0$ as the cumulative size of all the brackets excluding the last one. By construction,

$$U_4 = n_2 - 1,$$

so that n_2, the number of strings completed in round two, is the first unit in bracket five in round three. The marginal tax rate in each bin is then

$$\tau_j = \begin{cases} 0 & j = 1 \\ \frac{j-1}{10} & j = 2, ..., 9. \end{cases}$$

Finally, this means that the total tax liability is equal to

$$T_3^P(20qt_3) = \sum_{k=1}^{8} \tau_k \cdot 20 \cdot \max\{0, \min(w_k, q_3 \cdot t_3 - U_{k-1})\} + 0.8 \cdot 20 \cdot \max\{0, q_3 \cdot t_3 - U_8\},$$

Next, we want to derive an expression for the marginal tax rate in this setting. let $n_3 = q_3 \cdot t_3$ represent the number of strings completed in round three. The taxable amount in bracket k is

$$b_k(n) = \max\{0, \min(w_k, n_3 - U_{k-1})\}, \qquad k = 1, \ldots, 8.$$

Equivalently,

$$b_k(n_3) = \begin{cases} 0 & n_3 \leq U_{k-1}, \\ n_3 - U_{k-1} & U_{k-1} < n_3 < U_k, \\ w_k & n_3 \geq U_k. \end{cases}$$

Because the tax schedule is a step function, the derivative is not defined at bracket thresholds. We therefore use the right-hand derivative, which captures the tax applied to the next string completed. The right-hand derivative of $b_k(n_3)$ with respect to the number of strings completed is

$$b_k^+(n) = \begin{cases} 0 & n_3 < U_{k-1}, \\ 1 & U_{k-1} \leq n_3 < U_k, \\ 0 & n_3 \geq U_k. \end{cases}$$

Now taking the right-hand derivative of the progressive tax function with respect to the number of strings yields

$$\frac{d^+T_3^P(20n_3)}{dn_3} = \sum_{k=1}^{8} \tau_k \cdot 20 \cdot b_k^+(n_3) + 0.8 \cdot 20 \cdot \mathbb{1}\{n_3 \geq U_8\}.$$

Dividing by 20 gives the marginal tax rate per additional dollar earned:

$$MTR_3^P(n_3) = \sum_{k=1}^{8} \tau_k \mathbb{1}\{U_{k-1} \leq n_3 < U_k\} + 0.8 \cdot \mathbb{1}\{n_3 \geq U_8\}.$$

For any $n_3 < U_8$, exactly one bracket indicator equals one. Therefore the marginal tax rate is the marginal tax rate associated with the bracket containing n. For any $n_3 \geq U_8$, the marginal tax rate is 0.8. Hence,

$$MTR_3^P(n_3) = \begin{cases} \frac{k-1}{10} & \text{if } U_{k-1} \leq n_3 < U_k, \quad k = 1, \ldots, 8, \\ 0.8 & \text{if } n_3 \geq U_8. \end{cases}$$

Finally, the condition characterizing the optimal labour supply in this setting is

$$u'(c(t_3)) \left[20q_3(1 - MTR_3^P(q_3t_3)) - 1\right] = \nu'(t_3), \tag{4.7}$$